Lesson 18 material: Complex Correlation

6.4: Complex Correlation

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Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Explain some reasons that researchers use complex correlational designs.
2. Create and interpret a correlation matrix.
3. Describe how researchers can use partial correlation and multiple regression to statistically control for third variables.

As we have already seen, researchers conduct correlational studies rather than experiments when they are interested in noncausal relationships or when they are interested in causal relationships but the independent variable cannot be manipulated for practical or ethical reasons. In this section, we look at some approaches to complex correlational research that involve measuring several variables and assessing the relationships among them.

Assessing Relationships Among Multiple Variables

Most complex correlational research involves measuring several variables—either binary or continuous—and then assessing the statistical relationships among them. For example, researchers Nathan Radcliffe and William Klein studied a sample of middle-aged adults to see how their level of optimism (measured by using a short questionnaire called the Life Orientation Test) relates to several other variables related to having a heart attack (Radcliffe & Klein, 2002)[1]. These included their health, their knowledge of heart attack risk factors, and their beliefs about their own risk of having a heart attack. They found that more optimistic participants were healthier (e.g., they exercised more and had lower blood pressure), knew about heart attack risk factors, and correctly believed their own risk to be lower than that of their peers.

In another example, Ernest Jouriles and his colleagues measured adolescents' experiences of physical and psychological relationship aggression and their psychological distress. Because measures of physical aggression (such as the Conflict in Adolescent Dating Relationships Inventory and the Relationship Violence Interview) often tend to result in highly skewed distributions, the researchers transformed their measures of physical aggression into a dichotomous (i.e., binary) measure (0 = did not occur, 1 = did occur). They did the same with their measures of psychological aggression and then measured the correlations among these variables, finding that adolescents who experienced physical aggression were moderately likely to also have experienced psychological aggression and that experiencing psychological aggression was related to symptoms of psychological distress. (Jouriles, Garrido, Rosenfield, & McDonald, 2009)[2]

This approach is often used to assess the validity of new psychological measures. For example, when John Cacioppo and Richard Petty created their Need for Cognition Scale—a measure of the extent to which people like to think and value thinking—they used it to measure the need for cognition for a large sample of college students, along with three other variables: intelligence, socially desirable responding (the tendency to give what one thinks is the "appropriate" response), and dogmatism (Caccioppo & Petty, 1982)[3]. The results of this study are summarized in Table , which is a **correlation matrix** showing the correlation (Pearson's *r*) between every possible pair of variables in the study. For example, the correlation between the need for cognition and intelligence was +.39, the correlation between intelligence and socially desirable responding was +.02, and so on. (Only half the matrix is filled in because the other half would contain exactly the same information. Also, because the correlation between a variable and itself is always +1.00, these values are replaced with dashes throughout the matrix.) In this case, the overall pattern of correlations was consistent with the researchers' ideas about how scores on the need for cognition should be related to these other constructs.

Table : Correlation Matrix Showing Correlations Among the Need for Cognition and Three Other Variables Based on Research by Cacioppo and Petty (1982)

Need for cognition	**Intelligence**	**Social desirability**	**Dogmatism**	
Need for cognition	—			
Intelligence	+.39	—		
Social desirability	+.08	+.02	—	
Dogmatism	−.27	−.23	+.03	—

Factor Analysis

When researchers study relationships among a large number of conceptually similar variables, they often use a complex statistical technique called **factor analysis**. In essence, factor analysis organizes the variables into a smaller number of clusters, such that they are strongly correlated within each cluster but weakly correlated between clusters. Each cluster is then interpreted as multiple measures of the same underlying construct. These underlying constructs are also called "factors." For example, when people perform a wide variety of mental tasks, factor analysis typically organizes them into two main factors—one that researchers interpret as mathematical intelligence (arithmetic, quantitative estimation, spatial reasoning, and so on) and another that they interpret as verbal intelligence (grammar, reading comprehension, vocabulary, and so on). The Big Five personality factors have been identified through factor analyses of people's scores on a large number of more specific traits. For example, measures of warmth, gregariousness, activity level, and positive emotions tend to be highly correlated with each other and are interpreted as representing the construct of extraversion. As a final example,

researchers Peter Rentfrow and Samuel Gosling asked more than 1,700 university students to rate how much they liked 14 different popular genres of music (Rentfrow & Gosling, 2008)[4]. They then submitted these 14 variables to a factor analysis, which identified four distinct factors. The researchers called them *Reflective and Complex* (blues, jazz, classical, and folk), *Intense and Rebellious* (rock, alternative, and heavy metal), *Upbeat and Conventional* (country, soundtrack, religious, pop), and *Energetic and Rhythmic* (rap/hip-hop, soul/funk, and electronica); see Table .

Table : Factor Loadings of the 14 Music Genres on Four Varimax-Rotated Principal Components. Based on Research by Rentfrow and Gosling (2003)

Music-preference dimension				
Genre	***Reflective and Complex***	***Intense and Rebellious***	***Upbeat and Conventional***	***Energetic and Rhythmic***
Blues	***.85***	.01	-.09	.12
Jazz	***.83***	.04	.07	.15
Classical	***.66***	.14	.02	-.13
Folk	***.64***	.09	.15	-.16
Rock	.17	***.85***	-.04	-.07
Alternative	.02	***.80***	.13	.04
Heavy metal	.07	***.75***	-.11	.04
Country	-.06	.05	***.72***	-.03
Sound tracks	.01	.04	***.70***	.17
Religious	.23	-.21	***.64***	-.01
Pop	-.20	.06	***.59***	*.45*
Rap/hip-hop	-.19	-.12	.17	***.79***
Soul/funk	.39	-.11	.11	***.69***
Electronica/dance	-.02	.15	-.01	***.60***
Note. N = 1,704. All factor loadings .40 or larger are in italics; the highest factor loadings for each dimension are listed in boldface type.				

Two additional points about factor analysis are worth making here. One is that factors are not categories. Factor analysis does not tell us that people are *either* extraverted *or* conscientious or that they like *either* "reflective and complex" music *or* "intense and rebellious" music. Instead, factors are constructs that operate independently of each other. So people who are high in extraversion might be high or low in conscientiousness, and people who like reflective and complex music might or might not also like intense and rebellious music. The second point is that factor analysis reveals only the underlying structure of the variables. It is up to researchers to interpret and label the factors and to explain the origin of that particular factor structure. For example, one reason that extraversion and the other Big Five operate as separate factors is that they appear to be controlled by different genes (Plomin, DeFries, McClean, & McGuffin, 2008)[5].

Exploring Causal Relationships

Another important use of complex correlational research is to explore possible causal relationships among variables. This might seem surprising given the oft-quoted saying that "correlation does not imply causation." It is true that correlational research cannot unambiguously establish that one variable causes another. Complex correlational research, however, can often be used to rule out other plausible interpretations. The primary way of doing this is through the **statistical control** of potential third variables. Instead of controlling these variables through random assignment or by holding them constant as in an experiment, the researcher instead measures them and includes them in the statistical analysis called **partial correlation**. Using this technique, researchers can examine the relationship between two variables, while statistically controlling for one or more potential third variables.

For example, assume a researcher was interested in the relationship between watching violent television shows and aggressive behavior but she was concerned that socioeconomic status (SES) might represent a third variable that is driving this relationship. In this case, she could conduct a study in which she measures the amount of violent television that participants watch in their everyday life, the number of acts of aggression that they have engaged in, and their SES. She could first examine the correlation between violent television viewing and aggression. Let's say she found a correlation of +.35, which would be considered a moderate sized positive correlation. Next, she could use partial correlation to reexamine this relationship after statistically controlling for SES. This technique would allow her to examine the relationship between the part of violent television viewing that is independent of SES and the part of aggressive behavior that is independent of SES. If she found that the partial correlation between violent television viewing and aggression while controlling for SES was +.34, that would suggest that the relationship between violent television viewing and aggression is largely independent of SES (i.e., SES is not a third variable driving this relationship). On the other hand, if she found that after statistically controlling for SES the correlation between violent television viewing and aggression dropped to +.03, then that would suggest that SES is indeed a third variable that is driving the relationship. If, however, she found that statistically controlling for SES reduced the magnitude of the correlation from +.35 to +.20, then this would suggest that SES accounts for some, but not all, of the relationship between television violence and aggression. It is important to note that while partial correlation provides an important tool for researchers to statistically control for third variables, researchers using this technique are still limited in their ability to arrive at causal conclusions because this technique does not take care of the directionality problem and there may be other third variables driving the relationship that the researcher did not consider and statistically control.

Regression

Once a relationship between two variables has been established, researchers can use that information to make predictions about the value of one variable given the value of another variable. For, instance, once we have established that there is a correlation between IQ and GPA we can use people's IQ scores to predict their GPA. Thus, while correlation coefficients can be used to describe the strength and direction of relationships between variables, **regression** is a statistical technique that allows researchers to predict one variable given another. Regression can also be used to describe more complex relationships between more than two variables. Typically the variable that is used to make the prediction is referred to as the **predictor variable** and the variable that is being predicted is called the **outcome variable or criterion variable.** This regression equation has the following general form:

$$Y = b_1 X_1$$

Y in this formula represents the person's predicted score on the outcome variable, b_1 represents the slope of the line depicting the relationship between two variables (or the regression weight), and X1 represents the person's score on the predictor variable. You can see that to predict a person's score on the outcome variable (Y), one simply needs to multiply their score on the predictor variable (X) by the regression weight (b_1)

While **simple regression** involves using one variable to predict another, **multiple regression** involves measuring several variables (X_1, X_2, X_3,...X_i), and using them to predict some outcome variable (Y). Multiple regression can also be used to simply describe the relationship between a single outcome variable (Y) and a set of predictor variables (X_1, X_2, X_3,...X_i). The result of a multiple regression analysis is an equation that expresses the outcome variable as an additive combination of the predictor variables. This regression equation has the following general form:

$$Y = b_1 X_1 + b_2 X_2 + b_3 X_3 + ... + b_i X_i$$

The regression weights (b_1, b_2, and so on) indicate how large a contribution a predictor variable makes, on average, to the prediction of the outcome variable. Specifically, they indicate how much the outcome variable changes for each one-unit change in the predictor variable.

The advantage of multiple regression is that it can show whether a predictor variable makes a contribution to an outcome variable *over and above* the contributions made by other predictor variables (i.e., it can be used to show whether a predictor variable is related to an outcome variable after statistically controlling for other predictor variables). As a hypothetical example, imagine that a researcher wants to know how income and health relate to happiness. This is tricky because income and health are themselves related to each other. Thus if people with greater incomes tend to be happier, then perhaps this is only because they tend to be healthier. Likewise, if people who are healthier tend to be happier, perhaps this is only because they tend to make more money. But a multiple regression analysis including both income and health as predictor variables would show whether each one makes a contribution to the prediction of happiness when the other is taken into account (when it is statistically controlled). In other words, multiple regression would allow the researcher to examine whether that part of income that is unrelated to health predicts or relates to happiness as well as whether that part of health that is unrelated to income predicts or relates to happiness. Research like this, by the way, has shown both income and health make extremely small contributions to happiness except in the case of severe poverty or illness (Diener, 2000[6]).

The examples discussed in this section only scratch the surface of how researchers use complex correlational research to explore possible causal relationships among variables. It is important to keep in mind, however, that purely correlational approaches cannot unambiguously establish that one variable causes another. The best they can do is show patterns of relationships that are consistent with some causal interpretations and inconsistent with others.

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